

Ensinger RADEL® R-5500 Polyphenylsulfone (PPSU)

Category : Polymer , Thermoplastic , Polyphenylsulfone (PPSU)

Material Notes:

RADEL® R is a Polyphenylsulfone produced by Solvay Advanced Polymers. RADEL® R is in the same performance category as products like Polysulfone and ULTEM® (PEI). RADEL® R has increased performance versus Polysulfone and ULTEM® in terms of temperature, impact strength and chemical resistance. RADEL® R has exceptional resistance to repeated steam autoclaving without loss of dimensional stability or physical properties. RADEL® R comes in both opaque and transparent grades. The transparent grade is R-5500 and the opaque grade is R-5100. High tensile strength Excellent heat resistance Very high resistance to environmental stress Great mechanical strength High dielectric strength and stability Low dissipation factor Very good machinability and finishing Available in opaque and transparent grades RADEL® R series products are targeted at a number of industries and applications. The initial target market is the medical industry. RADEL® R is used in applications for surgical tools and instruments because of its resistance to autoclave sterilization damage. A second market for RADEL® R is electronics because of its temperature resistance and dielectric properties. Information Provided by Ensinger Industries, Inc.

Order this product through the following link:

http://www.lookpolymers.com/polymer_Ensinger-RADEL-R-5500-Polyphenylsulfone-PPSU.php

Physical Properties	Metric	English	Comments
Specific Gravity	1.29 g/cc	1.29 g/cc	ASTM D792
Density	1.29 g/cc	0.0466 lb/in³	ASTM D792
Water Absorption	0.37 % @Temperature 22.8 °C, Time 86400 sec	0.37 % @Temperature 73.0 °F, Time 24.0 hour	ASTM D570
Water Absorption at Saturation	1.1 % @Temperature 22.8 °C	1.1 % @Temperature 73.0 °F	ASTM D570

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	123	123	ASTM D785
Tensile Strength, Yield	69.6 MPa @Temperature 22.8 °C	10100 psi @Temperature 73.0 °F	ASTM D638
Elongation at Break	60 % @Temperature 22.8 °C	60 % @Temperature 73.0 °F	ASTM D638
Tensile Modulus	2.34 GPa	340 ksi	ASTM D638
Flexural Strength	91.0 MPa @Temperature 22.8 °C	13200 psi @Temperature 73.0 °F	ASTM D790
	2.41 GPa	350 ksi	

Mechanical Properties	Metric @ Temperature 22.8 °C	English @ Temperature 73.0 °F	Comments ASTM D790
Compressive Strength	98.94 MPa	14350 psi	ASTM D695
Izod Impact, Notched	6.94 J/cm @ Temperature 22.8 °C	13.0 ft-lb/in @ Temperature 73.0 °F	ASTM D256

Thermal Properties	Metric	English	Comments
CTE, linear	30.6 µm/m-°C	17.0 µin/in-°F	ASTM D696
Specific Heat Capacity	1.13 J/g-°C	0.270 BTU/lb-°F	
Maximum Service Temperature, Air	182 °C	360 °F	Intermittent
Deflection Temperature at 0.46 MPa (66 psi)	220 °C	428 °F	ASTM D648
Deflection Temperature at 1.8 MPa (264 psi)	216 °C	420 °F	ASTM D648
Vicat Softening Point	218 °C	424 °F	
Flammability, UL94	V-0	V-0	

Electrical Properties	Metric	English	Comments
Volume Resistivity	1.00e+15 ohm-cm	1.00e+15 ohm-cm	ASTM D257
Dielectric Constant	3.44 @ Frequency 60.0 Hz, Temperature 22.8 °C	3.44 @ Frequency 60.0 Hz, Temperature 73.0 °F	50% RH; ASTM D150
Dielectric Strength	14.2 kV/mm	360 kV/in	ASTM D149

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